

Design and Development of an Unmanned Ground Vehicle with Serial Manipulator and its Control Mechanism

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Abstract—The use of robotic vehicles for doing various jobs has been significantly increases in the last decade. Industries as well as militaries around the world, use them to perform their day to day operations. In this paper, a computationally efficient, cost-effective and user-friendly Arduino based, control methodology is presented to control an Unmanned Ground Vehicle (UGV) and its serial manipulator. A transmitter made up of a gamepad, WiFi module, and Arduino as a main controller is used to send commands. These commands reside on a cloud server from where the receiver picks them up and activates its various peripherals to fulfill them. The designed control mechanism is implemented on a fabricated UGV for real-time performance evaluation. Field tests indicate the effectiveness of the proposed methodology.

Index Terms—Unmanned Ground Vehicle, DualShock-2 Gamepad, FPV Camera, GPS, ESP8266 WiFi module, Buck Regulators.

I. INTRODUCTION

An unmanned ground vehicle (UGV) is a vehicle which operates without an on-board human presence. Mostly used in the areas where it may be inconvenient, dangerous or impossible for a human to operate, a UGV uses sensors data to observe its surroundings and operate autonomously. Usually, the information from sensors is passed to a human operator residing at base station which will then operate the UGV wirelessly. UGVs are widely develop for both civilians (from acting as fire brigades, ambulances and police to assisted in wide variety of tasks such as harvesting, transportation and inspection etc.) and military (i.e anti-mines, bomb disposal, surveillance etc.) purposes [1].

The paper presents the design and development of an Arduino based control methodology for UGV and its serial manipulator. Using WiFi protocol for communication and equipped with first person view camera (FPV) along-with GPS for surveillance and navigation, the UGV is manually controlled with a DualShock-2 (DS-2) gamepad.

The rest of paper is organized as follow: Section II presents literature review. Section III discusses mechanical design. Section IV discusses electric circuitry, Mathematical analysis and control methodology is elaborated in section V and VI respectively. Video feedback and UGV's navigation is presented in section VII while section VIII and IX presents conclusions and future work respectively.

II. RELATED WORK

Several works have been done in the development and control of unmanned ground vehicles. A mine detecting robotic vehicle working on wireless communication is discussed in [2]. Solving the problem of landmines, the vehicle is equipped with metal detector circuitry and various sensors with STC89C52RC MCU behaving as a control unit. The robot used CAN BUS for data communication. An android based mobile robot in [3] is used for indoor health care. The robot is based on iRobot Create product and used sets of smartphones having android operating system for transmitting and receiving data packet and commands. Use of internet of things (IoT) is observed in [4] where the robot named as "smart housekeeper", equipped with crawler chassis structure and head lifting machinery and having Cortex M4 as the main controller, used WiFi to connect with router. A smartphone is then used to remotely control smart housekeeper. The use of joystick for control is presented in [5] where an FPGA based UGV is used. The UGV used Compact Rio as a control unit along with LabVIEW for control and navigation. Another mines detection robot is presented in [6] where a prototype of a WiFi controlled robot having Arduino as a main controller is discussed. The robot made use of GPS, path navigation, obstacle detection and avoidance algorithms for navigation and accurately identification of mines. Wireless control of robots is also observed in [7] and [8]. A unique feature in the control

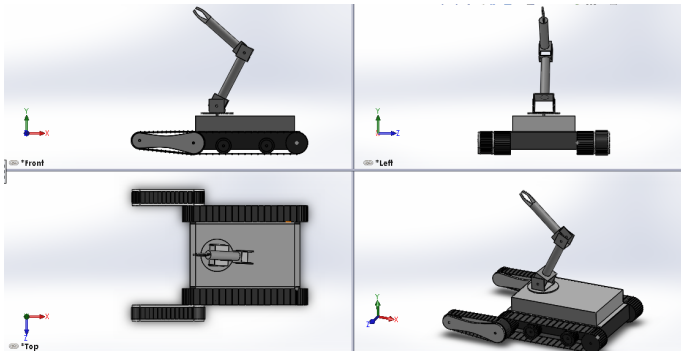


Fig. 1: CAD Design

system of [8] is observed where the control system is divided into two parts: the handheld remote control terminal based on Android smartphone and the on-board controller based on a micro-controller.

III. MECHANICAL DESIGN

The UGV which provides surveillance along with the ability to pick and place things mainly includes two parts, moving base which approach towards the target, and a manipulator to pick and place the object. The design and simulation of the real world robot in a three-dimensional environment is very crucial. Various computer-aided design software has been designed that help engineers to achieve this task efficiently. Therefore, SOLIDWORKS is used to design and simulate the UGV. Fig.1 shows the CAD design of the UGV's structure along with the manipulator.

A. Base Design

The UGV's base is made with stainless steel and aluminum plates, using chain-track locomotive mechanism. Each track is actuated by DC motor to drive wheels. To climb steep slopes and stairs easily, flappers are also attached to the base through ring and pinion differential mechanism. They also provide more traction and evenly distributes the weight of the UGV in rough and steep paths. TABLE 1 discusses the base design's specifications.

TABLE I: Base Design's Specifications

Material	Aluminium, Steel
Length with flapper extended	68 cm
Height	18 cm
Width	58 cm
Motors	x4
Drive Type	Differential

B. Serial Manipulator Design

The serial manipulator is fabricated using aluminum, and each joint is actuated by servo motor. The serial manipulator is of 4 degrees of freedom (DOF), with four motors (including gripper). It has a base rotation, shoulder, elbow, and wrist, which allows its movement in all three dimensions.

TABLE II: Manipulator Design's Specifications

Material	Aluminium
Degrees of Freedom	04
Payload	500 g
Maximum Horizontal Reach	64 cm
Maximum Vertical Reach	72 cm
Servo motors	x4

The actual design was fabricated using aluminum and stainless steel material to meet all design constraints as simulated in SOLIDWORKS. Fig.2 shows the complete hardware design model.

IV. ELECTRIC CIRCUITRY

Electric components required for controlling the UGV can be divided into two sides. The transmitting side (Transmitter) and the receiving side (Receiver).

A. Transmitter

A transmitter is designed to control the UGV remotely. It consists of:

1) *DualShock-2 Gamepad*: A Dualshock-2 gamepad is used to help the operator to control the UGV base and serial manipulator. The easy comprehension of gamepad by a naive makes it highly feasible for such operations. It connects with the main controller (i.e Arduino Mega) through SPI communication protocol.

2) *Arduino Mega - Controller*: Arduino Mega 2560 acts as a main controller. It receives a command from the operator, interpret them and activate necessary peripherals to transmit them.

3) *Wemos D1 Mini ESP8266 Dev Board*: Since Arduino Mega lacks a WIFI module so for transmission, Wemos D1 Mini ESP8266 Dev Board is used. It contains an integrated WiFi transceiver. On receiving the data packets from the main controller, the board sends them to the cloud, from where the receiver picks them up and act accordingly. Serial Communication protocol is used between Arduino Mega and WiFi board.



Fig. 2: Designed UGV

4) *Video Receiving Module*: Remote station also contains a video receiving module (operating on 5.8 GHz frequency) to receive the video output from the FPV camera. It can be easily connected to any monitor through HDMI cable for better visuals.

5) *Power Supply*: A variable power supply providing the output of 3.3V, 5V, and 7V is designed to energize the transmitting side.

B. Receiver

The receiving side, similar to the transmitter, contains "Wemos D1 Mini ESP8266 Dev Board" for fetching data from the cloud and "Arduino Mega" acting as a main controller for receiving, interpreting and activating the necessary output peripherals. Additional components in the receiving side are:

1) *Motor Driver Module*: Three IBT-2 H-bridge motor drivers are used to control the direction and speed of the DC motors independently.

2) *Buck Regulators*: The servo motor used in the shoulder joint of the serial manipulator has a stall torque of 38 kg.cm at 7.4 volts. To achieve the maximum efficiency, a linear regulator circuit using XL4015 IC is used to power it up. Remaining servo motors have a stall torque of 15 kg.cm at 6 volts. For them, a power circuit using LM2596 IC is designed.

3) *Batteries*: Two 12 volt 12Ah rechargeable lead-acid batteries are used to energize the receiving side.

4) *FPV Camera*: A wide angle FPV camera is installed at the serial manipulator for surveillance purposes.

5) *Video Transmission Module*: A video transmission module is used to wirelessly send the video output of the FPV camera at a transmission frequency of 5.8 GHz.

6) *GPS Module*: NEO- 6M GPS module is also used to track the location of UGV in real-time.

V. MATHEMATICAL ANALYSIS

Control of UGV (being a mobile system) requires a kinematic model. A differential drive model can be used for the designed UGV [9].

A. Mathematical Analysis for UGV's Base

Considering fig. 3, let
 L = Length of UGV tracks
 Θ = Angle of rotation
 R_W = UGV's wheels radius
 v_L, v_R = Linear velocities of left and right track respectively
 w = Rotational velocity of UGV
 R = Radius of rotation of UGV around the center

The linear velocity of right track is then given by:

$$v_R = R \times w \quad (1)$$

and for left track:

$$v_L = (R + L) \times w \quad (2)$$

Also the rotational velocity is given by:

$$w = \frac{2(v_L - v_R)}{L} \quad (3)$$

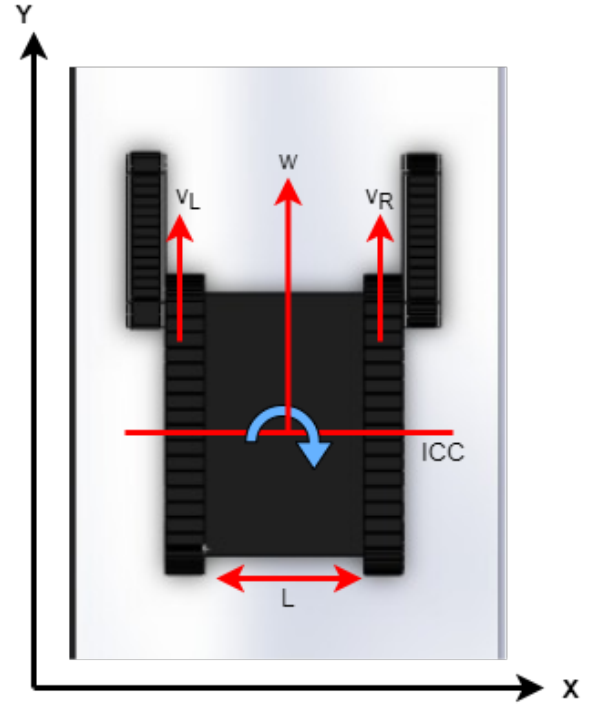


Fig. 3: Kinematics of UGV's Base

B. Mathematical Analysis for Manipulator

To pick and place an object on the ground, the shoulder and elbow joint of the manipulator need to be moved in some specific directions. Since servo motors are used at these joints, therefore, cartesian coordinates need to be converted into proper angles (forward kinematics) and vice versa (inverse kinematics) for servo motors to operate. Fig. 4 shows the reference image used for these conversations.

To equation for forward kinematics is:

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a_2 \cos(q_1 + q_2) & a_1 \cos(q_1) \\ a_2 \sin(q_1 + q_2) & a_1 \sin(q_1) \end{bmatrix}$$

Here

x, y = Cartesian coordinates

a_1, a_2 = Link's length 1 and 2 respectively

q_1, q_2 = Shoulder servo angle and elbow servo angle respectively.

For inverse kinematics:

$$q_1 = \arctan\left(\frac{y}{x}\right) - \arctan\left(\frac{a_2 \sin(q_2)}{a_1 + a_2 \cos(q_2)}\right) \quad (4)$$

$$q_2 = \arccos\left(\frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1 a_2}\right) \quad (5)$$

VI. CONTROL METHODOLOGY

To operate the UGV remotely, a wireless connection must be established between UGV's transmitter (i.e. base station) and its receiver (i.e. UGV itself). This necessary condition is

fulfilled by connecting both sides to WiFi (it must be noted that the network may differ for transmitter and receiver).

A. Transmitter

Transmission of commands to UGV occurs in the following manner:

1. Pressing of respective buttons of DualShock-2 gamepad by the operator.
2. Data transmission from DS-2 gamepad to Arduino Mega.
3. Data transmission from Arduino Mega to Wemos D1 Mini ESP8266 Dev Board.
4. Data transmission from Wemos D1 Mini ESP8266 Dev Board to Google Firebase (cloud server).

DS-2 gamepad is capable of sending 1 byte of data. Data is transmitted using two channels i.e. DATA and CMD. For synchronized transmission, a clock signal is also required by these channels. Another channel, known as ATT (attenuation) channel is also present, which will get high when some sort of signal is present on the CMD channel. For transmission from DS-2 to Arduino Mega, SPI Protocol is used.

On receiving data from the DS-2 gamepad, the Arduino Mega makes use of serial communication protocol to further transmit it to Wemos D1 Mini ESP8266 Dev Board. However, before transmission, data need to be converted into a specific format. For this purpose, a lightweight data - interchangeable format, Java-Script Object Notation (JSON) is used. It is easy for development boards like Arduino "to create" and "to parse" JSON structure. Similarly, for controllers, it's convenient to read and write JSON structures. Arduino Mega, after receiving data from DS-2, initialize a buffer of specific size and create a JSON structure. After that, the status of DS-2 buttons is fed in JSON root indexes. Finally, the root structure is written to the Wemos D1 board through serial communication.

Similarly, on receiving the data from Arduino, the Wemos board creates a buffer of the same size as created in Arduino. After parsing, the root indexes of the JSON structure are equalized to get a new data type or function. Finally, the data is sent to the cloud server (i.e. Google FireBase) which keeps record of the status of every button of DS-2.

B. Receiver

The commands received from the transmitter are processed by the receiver in the following manner:

1. Data capturing from cloud server by Wemos Board.
2. Data transmission from Wemos Board to Arduino Mega.
3. Arduino Mega, after comprehending the commands, generate necessary PWM to operate base and manipulator.

The data stored on the cloud is captured by the receiver's Wemos board. Stored data is of string format. Wemos converted it to integer datatype and sent it to Arduino through serial communication. The Arduino after receiving the data controls DC and servo motors by generating PWM following the statuses of datatypes.

Arduino Mega is connected with motor drivers (i.e. to DC motors) through logic level pins. The minimum control input level recognized by motor drivers is 3.3 volt. Servo motors are attached to PWM pins of Arduino. The phenomenon used to control the servo motor is known as Pulse Width Modulation (PWM). In this method, the period of the pulse is responsible for determining the angle of the servo motor.

VII. VIDEO FEEDBACK AND UGV NAVIGATION

As mentioned earlier, a wide-angle first-person view camera is attached to a manipulator that sends live video transmission to the base station at a frequency of 5.8 GHz. To aid the operator in knowing the condition of UGV, a GPS module is also attached to UGV. For navigation purposes, an android application is developed using Thinkable Online App Developer. The app displays UGV location on google maps with an option to zoom-in and zoom-out. Condition of UGV i.e. battery status, UGV's speed, etc. is also observed by tapping the "UGV" icon on the app.

VIII. CONCLUSION

An Arduino based, cost-effective and easily comprehensible, control mechanism is designed for UGV and its serial manipulator. The UGV can be operated remotely with DualShock - 2 gamepad using wireless communication. A GPS module and wide-angle FPV camera, attached with serial manipulator, provides real-time video feedback which helps in navigation and surveillance activities. The use of DualShock-2 gamepad makes the handling of UGV easy and smooth. Due to ease of use, it can be used by the military to carry out rescue/reconnaissance activities or industries for lifting, arranging and transporting their equipment.

The UGV can be made more advanced by adding more features to its control mechanism such as path planning or vision control algorithms. Adding various sensors with such algorithms can convert it into a fully autonomous self-driving UGV. Moreover, including algorithms like enemy tracking and various acoustic signature detection will make it highly useable for military operations.

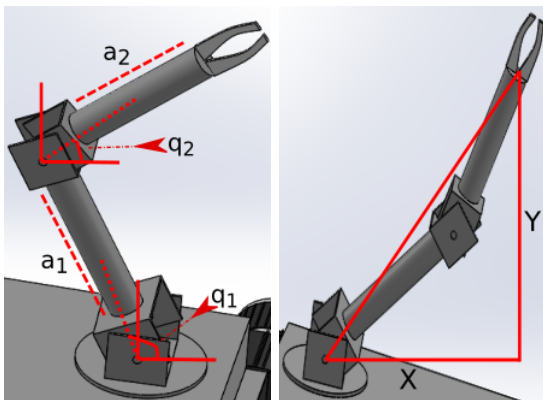


Fig. 4: Kinematics of UGV's Manipulator

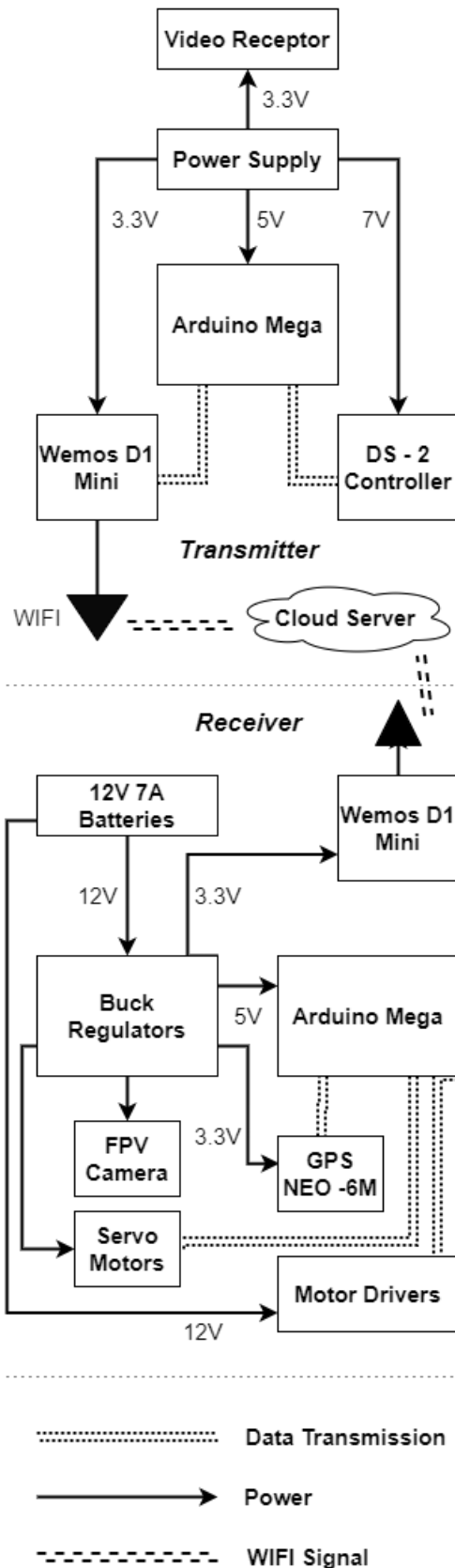


Fig. 5: Hardware Integration Diagram

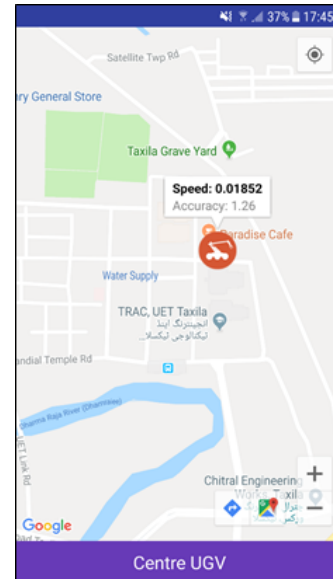


Fig. 6: UGV's Position on Google Map

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